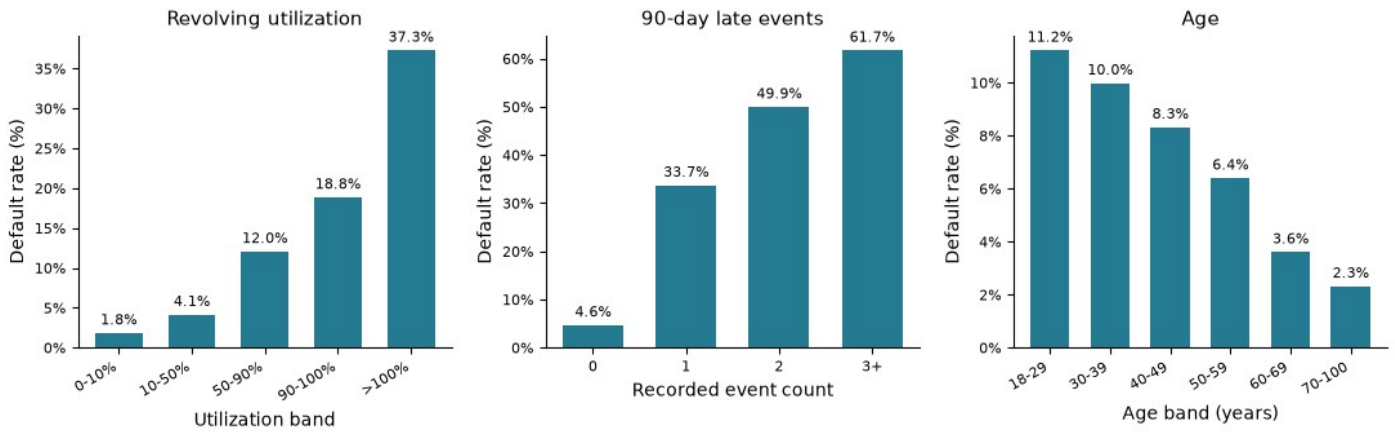
Credit Risk: Feature Distributions and Insights

Mehrdad Mohammadian | EDA

The analysis sample contains 9,878 defaults (6.6%). The three selected features have no missing values. Revolving utilization is strongly right-skewed. 7.3% of records are zero, while 768 values fall beyond the IQR outlier fence. Its extreme tail makes medians and grouped rates more informative than raw means.

Findings



- **Revolving utilization:** The default-group median is **0.830**, compared with **0.133** for non-defaults. The observed default rate rises from **1.8%** at 0-10% utilization to **18.8%** at 90-100%, and **37.3%** above 100%. High utilization can indicate limited remaining credit headroom.

- **90-day late payments:** This feature is zero-heavy (**94.6%** have no recorded event), so both class medians are zero. Group rates rise from **4.6%** with no late events to **33.7%** with one and **61.7%** with three or more.

- **Age:** The overall distribution is close to symmetric, but the default group is shifted younger (median age **46** versus **52** for non-defaults). Default rates decline from **11.2%** at ages 18-29 to **2.3%** at 70-100. The distributions still overlap, and age alone is not a reliable separator.

Final insight

I would start with `RevolvingUtilizationOfUnsecuredLines`, because 1) its median is 0.830 for defaults versus 0.133 for non-defaults, 2) it signals limited credit headroom, 3) it is associated with a higher default rate. I would add `NumberOfTimes90DaysLate`, because the default rate is much higher among customers with a recorded severe delay. My third choice is `age`, because the default-class median is 46 versus 52 years and the age bands show a consistent downward trend.

Note: These patterns show association, not causation; the report is exploratory and does not train a predictive model.